

Employment

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| Product Engineer Intern                                                                                                                                                                                                                                                                                                                                                                     | Isomer AI            | May 2025 – May 2026  |
| <ul style="list-style-type: none"><li>Ship 10+ production TypeScript features end-to-end: REST API integrations, real-time event pipelines (Pusher), multi-model AI agent tooling (Claude/GPT-4o/Gemini, MCP), Auth0 RBAC; 923 commits, 7 deployments.</li><li>Own backend stack (PostgreSQL, Redis, Docker, Terraform, Sentry); maintain CI/CD and resolve production incidents.</li></ul> |                      |                      |
| Graduate Research Assistant                                                                                                                                                                                                                                                                                                                                                                 | NC State University  | April 2025 – Present |
| <ul style="list-style-type: none"><li>Deploy genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets.</li><li>Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.</li></ul>                                                                                                 |                      |                      |
| Software Developer Intern                                                                                                                                                                                                                                                                                                                                                                   | Mphasis (HP Company) | Jun 2024 – Aug 2024  |
| <ul style="list-style-type: none"><li>Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%.</li><li>Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.</li></ul>                                                                                         |                      |                      |
| AI/ML Intern                                                                                                                                                                                                                                                                                                                                                                                | Capgemini            | Jun 2023 – Aug 2023  |
| <ul style="list-style-type: none"><li>Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%.</li><li>Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.</li></ul>                                                                                              |                      |                      |

Education

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| Raleigh, NC                                                                                                                                                                                                                     | North Carolina State University | Aug 2024 – May 2026 |
| <ul style="list-style-type: none"><li>Master of Science in Computer Science</li><li>Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.</li></ul> |                                 |                     |
| Mumbai, India                                                                                                                                                                                                                   | Mumbai University               | Aug 2020 – May 2024 |
| <ul style="list-style-type: none"><li>Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS &amp; Algorithms, Machine Learning, Blockchain.</li></ul>                        |                                 |                     |

Projects

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| PulseRank                                                                                                                                                                                                                                                                                                                                                                                  | Real-Time Leaderboard & Activity Feed | 2026 |
| <ul style="list-style-type: none"><li>Built event-driven leaderboard and activity feed in TypeScript with NestJS and Next.js: MySQL transactional writes, MongoDB feed projections with TTL, Redis sorted sets for sub-millisecond reads; SSE live updates.</li><li>Validated with Testcontainers; benchmarked hot paths with k6; OpenTelemetry + Prometheus; GitHub Actions CI.</li></ul> |                                       |      |
| CortexQ                                                                                                                                                                                                                                                                                                                                                                                    | Kubernetes LLM Inference Platform     | 2026 |
| <ul style="list-style-type: none"><li>Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing.</li><li>Instrumented OTel + Prometheus + Grafana observability; Helm + ArgoCD GitOps; k6 to 500 VUs; 88 pytest tests.</li></ul>              |                                       |      |
| AlgoPulse                                                                                                                                                                                                                                                                                                                                                                                  | Agentic AI Platform                   | 2025 |
| <ul style="list-style-type: none"><li>Built agentic AI orchestration platform on Temporal.io and Kafka supporting multi-step reasoning pipelines; sustained 2,000 async inference requests per second under sustained production load.</li><li>Replaced API polling with Supabase Edge Functions for real-time AI streaming; eliminated state loss under peak traffic.</li></ul>           |                                       |      |
| LLM Eval Harness                                                                                                                                                                                                                                                                                                                                                                           | Independent Project                   | 2025 |
| <ul style="list-style-type: none"><li>Built vendor-neutral LLM evaluation harness in Python comparing Claude, GPT-4o, and Gemini across 2,000 frozen labeled examples spanning factual recall, reasoning, and code generation tasks.</li><li>Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap CIs and McNemar tests across 6 metrics.</li></ul>               |                                       |      |

Languages and Technologies

- Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash
- Frontend & Full-Stack:** React, Next.js, NestJS, Node.js, REST APIs, SSE, WebSockets
- AI & Agents:** Claude/GPT-4o/Gemini APIs, MCP, Temporal.io, RAG, LLM evaluation, prompt engineering
- Cloud & Backend:** AWS, GCP, Docker, Kubernetes, PostgreSQL, Redis, Kafka, Terraform, GitHub Actions

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog